

# Aryan Pandey

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## EDUCATION

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|-------------------------------------------------------------------------------------|---------------------------------------|
| <b>University at Buffalo, State University of New York</b>                          | Buffalo, NY                           |
| <i>Master of Science (M.S.) in Computer Science and Engineering, Research Track</i> | <i>Jan 2025 – Dec 2026 (Expected)</i> |
| <b>University at Buffalo, State University of New York</b>                          | Buffalo, NY                           |
| <i>Bachelor of Science (B.S.) in Computer Engineering (GPA: 3.7, Dean's List)</i>   | <i>Aug 2018 – May 2022</i>            |

## EXPERIENCE

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| <b>Research Software Engineering Fellow</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Jun 2026 – Aug 2026  |
| <i>Princeton University – Princeton Neuroscience Institute</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Princeton, NJ</i> |
| <ul style="list-style-type: none"><li>Enabled researchers to fit cognitive models to behavioral data across multinode HPC (SLURM) clusters instead of a single machine, by building distributed systems infrastructure for parameter estimation in <i>PsyNeuLink</i>, an open-source simulation library.</li><li>Designed a Dask-based optimization backend that spins up a worker cluster inside a SLURM allocation, with near-linear throughput scaling to <b>128+ cores</b> and reproducible recovery of the same parameters as a serial fit.</li><li>Extended the backend to hierarchical multi-subject fitting, scaling population-level studies by running each subject's fit as an independent task with models cached per worker to bound memory.</li><li>Verified end-to-end parameter recovery on multi-node runs against an established Bayesian package (HSSM) to confirm statistical correctness.</li></ul>                                                                                                            |                      |
| <b>Software Engineer</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Jun 2022 – Apr 2024  |
| <i>Axon</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <i>Remote</i>        |
| <ul style="list-style-type: none"><li>Supported development of a large-scale archival system over one of the largest Azure databases in production, using Temporal and Azure Blob Storage to cut cloud storage spend by <b>\$1.5M/month</b>.</li><li>Improved evidence-video ingestion throughput by roughly <b>1.5×</b>, enabling the service to scale to <b>250 PB</b> while holding operational SLOs.</li><li>Held on-call ownership for globally deployed ingestion of body-cam, TASER, and CJIS data, maintaining Dockerized services at <b>99.99%+</b> availability.</li><li>Built Scala microservices for back-end infrastructure in a large production codebase, with unit testing, integration testing, and load testing across all services.</li><li>Deployed and released monthly updates across every production region via Kubernetes and Puppet.</li><li>Improved observability across back-end services with Grafana dashboards and restructured Splunk logging, reducing time-to-detect during incidents.</li></ul> |                      |
| <b>Teaching Assistant – VLSI, Computer Architecture, Computer Security</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Aug 2025 – Present   |
| <i>University at Buffalo</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <i>Buffalo, NY</i>   |
| <ul style="list-style-type: none"><li>Led labs on CMOS logic and standard-cell design: inverter sizing, NAND/NOR cells, ring oscillators, and SRAM bitcells.</li><li>Taught Cadence Virtuoso schematic, simulation, and layout workflows including parasitic extraction and DRC/LVS sign-off; guided students through timing/power trade-offs, drive-strength selection, and floorplanning for small ASIC blocks.</li><li>Mentored teams through full-flow hierarchical mixed-signal IC projects (DPLLs, TRNGs) from spec to tapeout, reviewing iterations and debugging design and verification issues.</li><li>Authored lab notes, debugging checklists, assignments, and exams; held weekly office hours and ran review sessions across combined undergraduate/graduate sections.</li></ul>                                                                                                                                                                                                                                      |                      |
| <b>Teaching Assistant – Computer Security, Computer Organization</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Aug 2020 – Dec 2021  |
| <i>University at Buffalo</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <i>Buffalo, NY</i>   |
| <ul style="list-style-type: none"><li>Ran weekly recitations on CPU architecture, ALU implementation, binary/hex computation, and pipelining; led exam review sessions.</li><li>Covered secure design, malware, risk management, authentication, OS and network security, cryptography, and social engineering; demonstrated common vulnerabilities and their countermeasures.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                      |
| <b>Software Engineer – Research Internship</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Jan 2020 – Aug 2020  |
| <i>University at Buffalo</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <i>Buffalo, NY</i>   |
| <ul style="list-style-type: none"><li>Built the C back-end for a portable FASTQ/A DNA analysis machine, including an inotify-based watchdog that detected and routed new files into the analysis pipeline asynchronously.</li><li>Parallelized the pipeline with OpenMP across all cores for an <b>8×</b> speedup over the single-threaded baseline.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                      |

## PROJECTS

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### Recurrent Differentiable Weightless Networks (RDWN) | *PyTorch, Verilog, Vivado* Feb 2026 – Present

- Added recurrent feedback paths to Differentiable Weightless Networks, a LUT-only architecture that replaces multiply-accumulates with table lookups for orders-of-magnitude cheaper FPGA inference than quantized DNNs; each trained lookup table maps directly onto a physical FPGA LUT-6.
- On Spiking Heidelberg Digits, the recurrent variant reached **82.2%** test accuracy vs. **67%** for the feedforward baseline; **97.75%** on N-MNIST.
- Wrote a PyTorch → Verilog export tool that extracts trained truth tables and learned wiring into synthesizable RTL; demonstrated the trained model end-to-end on FPGA at **0.6W**.

### ResearchShelf | *Python, FastAPI, Vue 3, SQLite, Docker, Claude Code* Jul 2026 – Present

- Built and released a self-hosted, multi-user research-paper platform with role-based team libraries, background processing, REST APIs, and a responsive browser-based PDF reader.
- Engineered PDF ingestion across Zotero, DOI/arXiv, Crossref, OpenAlex, and Unpaywall with SHA-256 deduplication and FTS5 full-text search; exposed authenticated workflows to LLM agents via MCP.
- Published to PyPI with automated GitHub releases and pytest/web-build CI covering **29 test modules across 6 OS/Python configurations** (Linux, macOS, Windows; Python 3.11/3.12).

### CUDA Neural Network Optimization for On-the-Fly Training | *CUDA, PyTorch, Triton* Mar 2025 – May 2025

- Extended Hugging Face training to reduce host-device transfers: kept generated samples on the GPU and transferred them only on mismatch, cutting end-to-end training time by roughly **20%**.
- Overlapped H2D transfers, D2H transfers, and computation using pinned memory and multi-stream execution; reduced a stress-test step from **46.71s to 39.34s**.
- Profiled the pipeline to locate data-movement stalls and GPU-idle regions; improved utilization through stream scheduling and batching changes.

### RL-Based Multi-Objective Neural Architecture Search for SNNs | *PyTorch, REINFORCE* Jan 2026 – Present

- Built an LSTM-controller NAS pipeline (REINFORCE with baseline) that searches spiking network architectures against a joint objective of accuracy, spike sparsity, and temporal coding quality.
- Designed a gated reward that blocks degenerate solutions where the search converges on networks that never spike, the dominant failure mode for multi-objective SNN search.
- On Spiking Heidelberg Digits, the controller converged to a single architecture and outperformed random search under a matched compute budget.

### SRAM Physically Unclonable Function | *Cadence Virtuoso, GPDK045* Aug 2024 – Dec 2024

- Designed a 6T-SRAM array exploiting charge-sharing and process variation for PUF-based cryptographic key generation; built transistor-level schematics and layouts in 45nm.
- Characterized bit stability and readout margins via Monte Carlo simulation and post-layout parasitic extraction.

## TECHNICAL SKILLS

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**Languages:** Python, C/C++, CUDA, Scala, Java, SQL, JavaScript, Verilog, Bash, ARM/x86/MIPS Assembly

**Machine Learning & Scientific Computing:** PyTorch, Hugging Face, Triton, Norse, NumPy/SciPy, Dask, OpenMP, Nsight profiling

**Backend & Data:** FastAPI, SQLAlchemy, REST APIs, gRPC/Protobuf, MCP, SQLite/FTS5, Temporal, Kafka, Spark/Databricks, Cassandra, Vue 3, pytest, unit/integration/load testing

**Cloud & Infrastructure:** Docker, Kubernetes, Azure, Terraform, SLURM, Linux, Git, GitHub Actions, CI/CD, Prometheus/Grafana, Splunk

**Hardware/EDA:** Xilinx Vivado, Cadence Virtuoso, SPICE, DRC/LVS, parasitic extraction